

Problem 1.8

Using compound interest formula, what principal does Andrew need to invest at 15% compounding annually so that he ends up with \$10,000 at the end of five years?

Solution

Using the compound interest formula

$$A = P(1 + i)^t,$$

we substitute

$$A = 10,000, \quad i = 0.15, \quad t = 5.$$

Then,

$$10,000 = P(1.15)^5.$$

Solving for P ,

$$P = \frac{10,000}{(1.15)^5}.$$

Calculating,

$$P \approx \frac{10,000}{2.0113571875} \approx 4,971.77.$$

Hence, Andrew needs to invest approximately

$$\boxed{\$4,971.77}.$$